Computer Project 2: Euler Equation Solver
Handed out 25th October, due by 7th November

Write a computer program to solve the Euler equations (1) using finite-volume methods.

$$\frac{\partial \vec{q}}{\partial t} + \frac{\partial \vec{F}}{\partial x} = 0 \tag{1}$$

The extents of your 1-D domain are from $x = -1$ to $x = 1$ with 40 intervals. The initial condition consists of a single discontinuity at x = 0, with uniform conditions on either side of it. Specifically, the following set of test cases are to be considered:

- $[\rho\ u\ T] = [1.0\ 0.0\ 500.0]$; $x \leq 0$; $[\rho\ u\ T] = [2.5\ 0.0\ 200.0]$; $x > 0$;
- A stationary compression shock at $x = 0$ with $P = 100$ Pa, $M = 3.0$, and $T = 401.8$ K to the left of $x = 0$. Use normal shock relations in your program to calculate the state for $x > 0$ for initialising the solution.
- $[\rho\ u\ T] = [2.5\ 0.0\ 500.0]$; $x \leq 0$; $[\rho\ u\ T] = [1.0\ 0.0\ 250.0]$; $x > 0$;

The working fluid can be assumed to be air. Close your system of equations with the assumption that air is an ideal gas. Assume $R = 287.0$ J/kg $\cdot$ K for air.

Use double precision for your program.

Your code should construct the flux at the interfaces using the following schemes:

- Steger-Warming scheme
- van Leer scheme (with and without a common interface sound speed)
- AUSM (use common interface sound speed)
- Roe scheme

Please assume a piecewise constant distribution for the variables within cells. **Your code should have capabilities to use Euler explicit time integration and a 4th-order explicit Runge-Kutta time integration.**

Your output should include ϕ vs. x plots, where $\phi \in \{\rho, u, P, T\}$, for your choice of total time. Also show the exact solution in your plots.

You need to present your work in a technical presentation format.

- In your title page please write your names and roll numbers.
- Your results should include comparison of schemes for first two test cases for any time $t > 0$ using an Euler explicit time integration.
- Please present comparison of Euler explicit and RK time integration schemes for the third test case using any flux scheme.
- Please submit a pdf copy of your presentation with the following naming convention: AS5460-<ROLL>-Project-Final.pdf where 'ROLL' refers to your roll number(s).
- **I would expect each student to write their own code. No sharing of code is acceptable between students and absolutely no collaboration on the reports is permitted. This can result in getting a U grade in this course.**